Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	**TE**	**Semester:**	**V**
Course Code:	**CSC502**	**Course Name:**	**WC**

Name of Student:	**Pranita Kumbhar**
Roll No. :	**70**
Experiment No.:	**04**
Title of the Experiment:	**To implement Javascript basics.**
Date of Performance:	**06/08/25**
Date of Submission:	**13/08/25**

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Ms. Kshitija Gharat

Signature :

Date:

Aim - To implement Javascript basics

Objective -

1. The objective of this experiment is to understand the basic function used in Javascript
2. Understand how to use the different basic functions for creating web applications.

Requirement -

1. An editor (Visual Studio Code)
2. Browser (Google Chrome)

Theory -

JavaScript is a lightweight, cross-platform, and interpreted compiled programming language which is also known as the scripting language for web pages. It is well-known for the development of web pages, many non-browser environments also use it. JavaScript can be used for Client-side developments as well as Server-side developments. Javascript is both an imperative and declarative type of language. JavaScript contains a standard library of objects, like Array, Date, and Math, and a core set of language elements like operators, control structures, and statements.

Code: Html code

Html code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  <h2>JavaScript variables</h2>
  <p id="demo"></p>
  <h2>Javascript Function</h2>
  <p id="demo1"></p>
  <h2>Javascript Condition(if..else)</h2>
  <p id="demo2"></p>
  <h2>Javascript loop(for loop)</h2>
  <p id="demo3"></p>
  <h2>JavaScript events</h2>
  <button onclick="document.getElementById('demo4').innerHTML=Date()">The time is?</button>
  <p id='demo4'></p>
  <h2>Javascript DOM CSS</h2>
  <h1 id="id1">My Heading 1</h1>
<button type="button" onclick="document.getElementById('id1').style.color = 'red'">Click Me!</button>
```

```
  <p id="p1">Hello World!</p>
  <p>The paragraph above was changed by a script.</p>
  <p>Please input a number between 1 and 10:</p>
  <input id="numb">
  <button type="button" onclick="myFunction()">Submit</button>
  <p id="demo5"></p>
  <script src="java1.js"></script>
</body>
</html>
```

Javascript code

```
var x = 5;
var y = 6;
var z = x + y;
document.getElementById("demo").innerHTML =
"The value of z is: " + z;
function myFunction(p1, p2) {
  return p1 * p2;
}
document.getElementById("demo1").innerHTML = "the value of function multiplication is:" +
myFunction(4, 3);
var a=4;
var b="";
if(a%2==0){
  b="even no";
}else{
  b="odd no";
}
document.getElementById("demo2").innerHTML = "the number is: " +b;
const cars = ["python", "java", "C", "C++", "HTML", "React"];
let text = "";
for (let i = 0; i < cars.length; i++) {
  text += cars[i] + "<br>";
}
document.getElementById("demo3").innerHTML = text;
document.getElementById("p1").innerHTML = "Kavya Naik"
function myFunction() {
  // Get the value of the input field with id="numb"
  let x = document.getElementById("numb").value;
  // If x is Not a Number or less than one or greater than 10
  let text;
  if (isNaN(x) || x < 1 || x > 10) {
          text = "Input not valid";
  } else {
```

```
        text = "Input OK";
  }
  document.getElementById("demo5").innerHTML = text;
}
```

Program-

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>JavaScript Basics Experiment</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Arial', sans-serif;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            min-height: 100vh;
            padding: 20px;
        }
        .container {
            max-width: 900px;
            margin: 0 auto;
            background-color: white;
            border-radius: 10px;
            padding: 30px;
            box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
        }
        h1 {
            text-align: center;
            color: #333;
            margin-bottom: 30px;
            border-bottom: 3px solid #667eea;
            padding-bottom: 10px;
        }
        .section {
            margin-bottom: 25px;
            padding: 20px;
            background-color: #f8f9fa;
            border-left: 4px solid #667eea;
            border-radius: 5px;
        }
        h2 {
```

```
    color: #667eea;
    font-size: 20px;
    margin-bottom: 10px;
}
p {
    color: #555;
    line-height: 1.6;
    margin: 8px 0;
}
button {
    background-color: #667eea;
    color: white;
    border: none;
    padding: 10px 20px;
    border-radius: 5px;
    cursor: pointer;
    font-size: 14px;
    margin: 5px 5px 5px 0;
    transition: 0.3s;
}
button:hover {
    background-color: #764ba2;
    transform: scale(1.05);
}
input {
    padding: 8px 12px;
    margin: 10px 0;
    border: 2px solid #ddd;
    border-radius: 5px;
    font-size: 14px;
    width: 200px;
}
input:focus {
    outline: none;
    border-color: #667eea;
    box-shadow: 0 0 5px rgba(102, 126, 234, 0.3);
}
.output {
    background-color: #e7f3ff;
    padding: 12px;
    margin: 10px 0;
    border-radius: 5px;
    color: #333;
    border-left: 3px solid #667eea;
    min-height: 30px;
}
.demo-time {
    font-size: 18px;
    font-weight: bold;
    color: #764ba2;
```

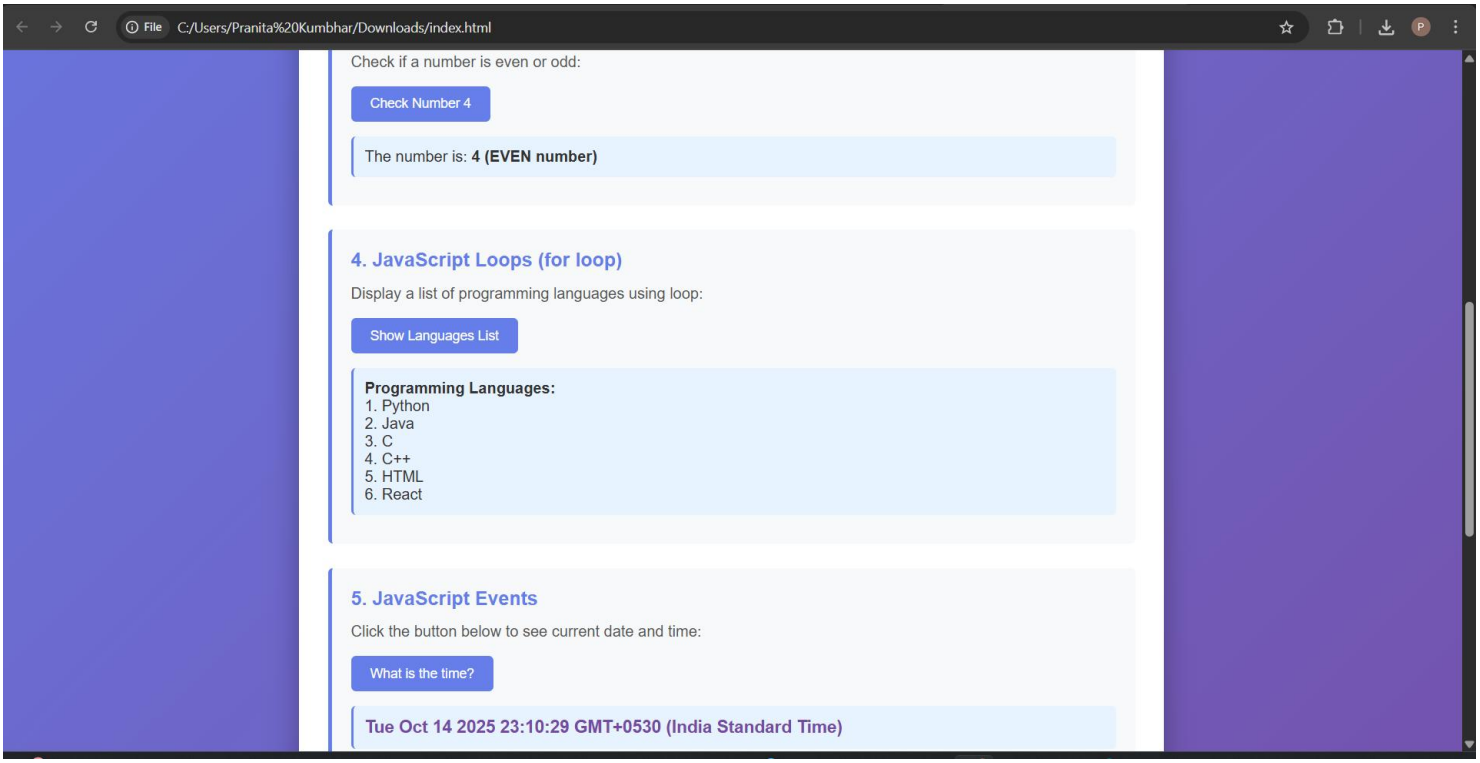
```
        }
    </style>
</head>
<body>

<div class="container">
    <h1> JavaScript Basics Experiment</h1>

    <!-- 1. Variables Section -->
    <div class="section">
        <h2>1. JavaScript Variables</h2>
        <p>Variables store data values. Here we add two variables:</p>
        <button onclick="showVariables()">Show Variables</button>
        <div id="demo" class="output"></div>
    </div>

    <!-- 2. Functions Section -->
    <div class="section">
        <h2>2. JavaScript Functions</h2>
        <p>Functions perform specific tasks. Here's a multiplication function:</p>
        <button onclick="showFunction()">Calculate Function (4 × 3)</button>
        <div id="demo1" class="output"></div>
    </div>

    <!-- 3. Conditions Section -->
    <div class="section">
        <h2>3. JavaScript Conditions (if...else)</h2>
        <p>Check if a number is even or odd:</p>
        <button onclick="showCondition()">Check Number 4</button>
        <div id="demo2" class="output"></div>
    </div>

    <!-- 4. Loops Section -->
    <div class="section">
        <h2>4. JavaScript Loops (for loop)</h2>
        <p>Display a list of programming languages using loop:</p>
        <button onclick="showLoop()">Show Languages List</button>
        <div id="demo3" class="output"></div>
    </div>

    <!-- 5. Events Section -->
    <div class="section">
        <h2>5. JavaScript Events</h2>
        <p>Click the button below to see current date and time:</p>
        <button onclick="showCurrentTime()">What is the time?</button>
        <div id="demo4" class="output"></div>
    </div>

    <!-- 6. DOM CSS Section -->
    <div class="section">
```

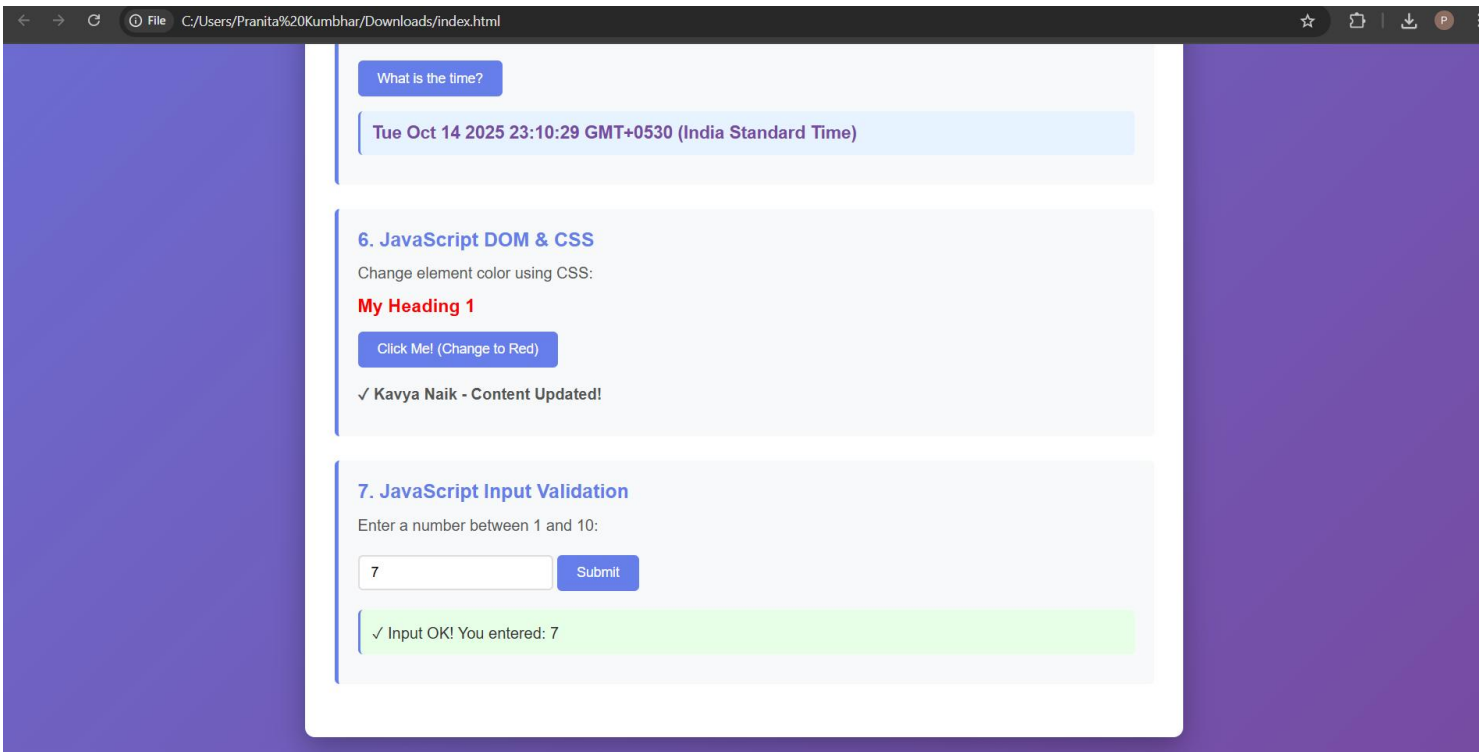
```
        <h2>6. JavaScript DOM & CSS</h2>
        <p>Change element color using CSS:</p>
        <h3 id="id1" style="color: black; margin: 10px 0;">My Heading 1</h3>
        <button onclick="changeColor()">Click Me! (Change to Red)</button>
        <p id="p1" style="margin: 10px 0; font-weight: bold;"></p>
    </div>

    <!-- 7. Input Validation Section -->
    <div class="section">
        <h2>7. JavaScript Input Validation</h2>
        <p>Enter a number between 1 and 10:</p>
        <input type="number" id="numb" placeholder="Enter number (1-10)">
        <button onclick="validateInput()">Submit</button>
        <div id="demo5" class="output"></div>
    </div>
</div>

<!-- JavaScript Code (Embedded) -->
<script>
    // 1. Variables Demo
    function showVariables() {
        var x = 5;
        var y = 6;
        var z = x + y;
        document.getElementById("demo").innerHTML =
            "x = " + x + "<br>y = " + y + "<br><strong>z (x + y) = " + z + "</strong>";
    }

    // 2. Function Demo
    function myFunction(p1, p2) {
        return p1 * p2;
    }

    function showFunction() {
        document.getElementById("demo1").innerHTML =
            "Function: myFunction(4, 3)<br><strong>Result (4 × 3) = " + myFunction(4, 3) + "</strong>";
    }

    // 3. Condition Demo
    function showCondition() {
        var a = 4;
        var b = "";
        if (a % 2 == 0) {
            b = "EVEN number";
        } else {
            b = "ODD number";
        }
        document.getElementById("demo2").innerHTML =
            "The number is: <strong>" + a + " (" + b + ")</strong>";
    }
```

```
    // 4. Loop Demo
    function showLoop() {
        const cars = ["Python", "Java", "C", "C++", "HTML", "React"];
        let text = "<strong>Programming Languages:</strong><br>";
        for (let i = 0; i < cars.length; i++) {
            text += (i + 1) + ". " + cars[i] + "<br>";
        }
        document.getElementById("demo3").innerHTML = text;
    }

    // 5. Event Demo
    function showCurrentTime() {
        document.getElementById("demo4").innerHTML =
            "<span class='demo-time'>" + new Date() + "</span>";
    }

    // 6. DOM CSS Demo
    function changeColor() {
        document.getElementById("id1").style.color = "red";
        document.getElementById("p1").innerHTML = "✓ Kavya Naik - Content Updated!";
    }

    // 7. Input Validation Demo
    function validateInput() {
        let x = document.getElementById("numb").value;
        let text;
        if (isNaN(x) || x < 1 || x > 10) {
            text = " Input NOT valid! Please enter a number between 1 and 10.";
            document.getElementById("demo5").style.background = "#ffe7e7";
        } else {
            text = "✓ Input OK! You entered: " + x;
            document.getElementById("demo5").style.background = "#e7ffe7";
        }
        document.getElementById("demo5").innerHTML = text;
    }
</script>

</body>
</html>
```

Output:

File C:/Users/Pranita%20Kumbhar/Downloads/index.html
🚀 JavaScript Basics Experiment
1. JavaScript Variables
Variables store data values. Here we add two variables:
Show Variables
x = 5
y = 6
z (x + y) = 11
2. JavaScript Functions
Functions perform specific tasks. Here's a multiplication function:
Calculate Function (4 × 3)
Function: myFunction(4, 3)
Result (4 × 3) = 12
3. JavaScript Conditions (if...else)


File C:/Users/Pranita%20Kumbhar/Downloads/index.html
Check if a number is even or odd:
Check Number 4
The number is: 4 (EVEN number)
4. JavaScript Loops (for loop)
Display a list of programming languages using loop:
Show Languages List
Programming Languages:
1. Python
2. Java
3. C
4. C++
5. HTML
6. React
5. JavaScript Events
Click the button below to see current date and time:
What is the time?
Tue Oct 14 2025 23:10:29 GMT+0530 (India Standard Time)



Conclusion - Javascript is a lightweight, interpreted, object-oriented language with first-class functions, and is best known as the scripting language for Web pages.